

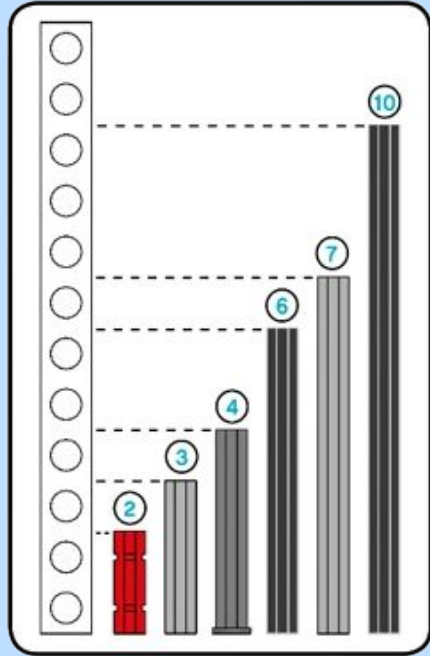
LESSON 3

Intro to Spike Prime

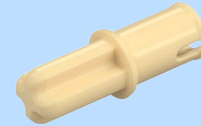


Spike Prime Materials - Axle

Axle



Pin/Axle Connector
2 Types
Friction/Frictionless(Beige)



Spike Prime Materials - Axle Connectors



Lego Technic Axle Building Tips

Transfer Motion & Power

Axles carry rotation from one part to another, like from a motor to wheels, gears, arms, or pulleys.

Allow Parts to Spin

Axles act as the center point for wheels, gears, levers, catapults, and moving arms.



Maintain Alignment

Axles keep gears, wheels, and beams lined up so the build stays stable and moves correctly.



Balloon Propulsion & Newton's Third Law

Question: What makes a balloon fly around the room when you let it go?

The Explanation:

- The air pushes backward out of the nozzle.
- The balloon pushes the air backward.
- The air pushes the balloon forward.



This is:

Newton's Third Law

"For every action, there is an equal and opposite reaction."

Today's Building Challenge

Project: Balloon Powered Car

Phase 1: Build & Iteration

- Follow initial building instructions
- Modify and add parts to improve performance

Phase 2: Performance Trials

Challenge 1: Max Distance

How far can your car travel?

Challenge 2: Sprint Speed

How fast can your car travel 1 foot?